the image and the term resolving power is associated with the ability of the optical instrument. Resolution and resolving power are reciprocal of each other.

EXAMPLE 7.14

The optical telescope in the Vainu Bappu observatory at Kavalur has an objective lens of diameter 2.3 m. What is its angular resolution if the wavelength of light used is 589 nm?

Solution

$a = 2.3$ m; $\lambda = 589$ nm $= 589\times10^{-9}$ m; $\theta = ?$

The equation for angular resolution is,

$$\theta = \frac{1.22\lambda}{a}$$

Substituting,

$$\theta = \frac{1.22\times589\times10^{-9}}{2.3} = 3.124\times10^{-7}$$

$$\theta = 3.124\times10^{-7}\,\text{rad} \;\; (\text{or}) \;\; \theta = 0.0011'$$

Note: The angular resolution of human eye is approximately, 3×10^{-4} rad $\approx 1.03'$.

7.5 POLARISATION

Both, longitudinal and transverse waves exhibit the phenomena of interference and diffraction. In fact, even sound waves demeonstrate the above two phenomenon. Since light is an electromagnetic wave, it is transverse in nature. The transverse nature of light wave is proved in the phenomenon called polarisation. **The phenomenon of restricting the vibrations of light (electric or magnetic field vectors) to any one direction perpendicular to the direction of propagation of wave is called *polarisation* of light**. In this lesson the electric field is only considered for discussion.

7.5.1 Plane polarised light

An *unpolarised light* is a transverse wave which has vibrations in all directions in a plane perpendicular to the direction of propagation of wave as shown in Figure 7.25(a). All these vibrations could be resolved into two normal components as shown in Figure 7.25(b), which still represents unpolarised light. **If the vibrations of a wave are present in only one direction in a plane perpendicular to the direction of propagation, then the light is said to be *polarised* (or) *plane polarised light*** as shown in Figure 7.25(c) and 7.25(d).

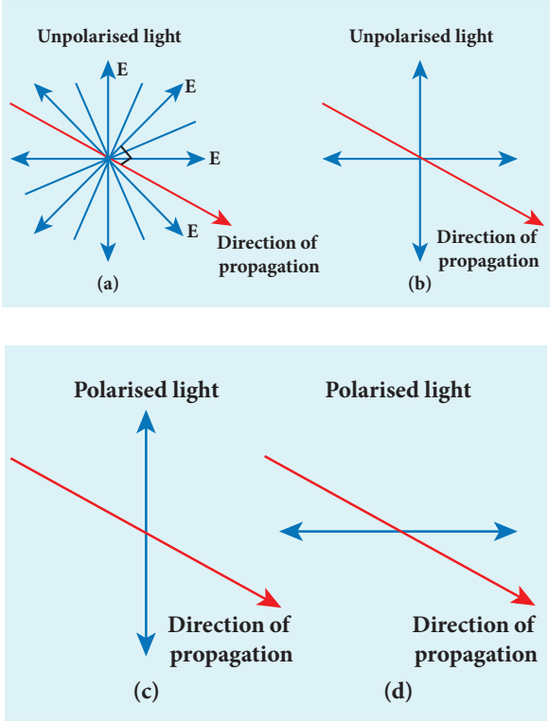


Figure 7.25 Unpolarised and polarised light

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